

PROJECT BASED LEARNING (PBL) PROPOSAL

Shopping Mall Sales Management System

A Role-Based, Multi-Counter Desktop Application for Real-Time Retail Sales, Inventory, and Business Intelligence

JAVA · MYSQL · MVC ARCHITECTURE · POS BILLING

TEAM RECURSION

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01 Executive Summary

Retail counters inside shopping malls still rely heavily on manual registers, spreadsheets, or disconnected billing tools that were never designed for a multi-shop environment. This proposal presents the design and development plan for a Shopping Mall Sales Management System (SMSMS) — a Java Swing desktop application backed by a MySQL database — that unifies point-of-sale billing, inventory control, staff management, and business analytics into a single, role-secured platform built specifically around how a mall actually operates: many shops, one supervising authority, and a constant need for accurate, real-time stock and revenue visibility.

Unlike typical single-shop billing software, this system is architected from the ground up for a multi-counter, multi-role environment, with a layered MVC design, transactional data integrity, and a reporting engine that turns raw sales data into actionable insight for mall management. The result is a project that is academically rigorous — covering software architecture, database design, and applied UI engineering — while solving a genuine, everyday operational problem.

In one line: A secure, offline-capable, multi-shop POS and inventory platform that gives mall management the same real-time visibility a large retail chain would pay enterprise software for.

02 Background & Motivation

Small and mid-sized shopping malls typically house dozens of independently operating shops, each handling billing and stock differently — some on paper, some on generic calculator-style POS software with no shared reporting layer. This creates three recurring problems for mall management: no unified view of total mall revenue, no early warning on stock-outs across shops, and no reliable audit trail when discrepancies occur.

This project is motivated by the gap between what commercial enterprise retail suites offer (often expensive, cloud-dependent, and overbuilt for a single mall) and what most malls actually use today (fragmented, manual, or spreadsheet-based). The goal is to design a right-sized, locally-hosted system that a mall's IT setup can realistically own and run.

03 Problem Statement

- **Fragmented billing:** Each shop/counter bills independently with no centralized transaction record for the mall as a whole.
- **Manual inventory:** Stock counts are tracked on paper or basic spreadsheets, leading to stock-outs, overstocking, and shrinkage that goes unnoticed.
- **No role separation:** Cashiers, shop managers, and mall administrators typically share the same access level, creating accountability and security gaps.
- **Zero real-time insight:** Management has no dashboard to see daily sales, best-selling categories, or underperforming counters without manually collecting data from each shop.
- **Error-prone returns/refunds:** Without a linked invoice history, returns are handled ad hoc, making stock and cash records unreliable.

04 Aim & Objectives

Aim: To design and implement a secure, role-based desktop application that centralizes sales, inventory, and reporting for a multi-shop mall environment using Java Swing and MySQL.

Specific Objectives

- ✓ Design a normalized MySQL schema that supports multi-shop products, sales, purchases, and returns with full referential integrity.
- ✓ Implement a transactional POS billing module that never lets a sale complete without a matching, atomic stock deduction.
- ✓ Build role-based access control distinguishing Admin, Manager, and Cashier privileges at both the UI and data layer.
- ✓ Develop a real-time analytics dashboard (sales trends, top products, low-stock alerts) using JFreeChart.
- ✓ Provide exportable PDF/Excel reports for management review and audit purposes.
- ✓ Package the system as a standalone, installable desktop application requiring no internet dependency.

05 What Makes This Project Unique

Most student POS projects stop at “add product, sell product, print bill.” This proposal is scoped to go further, in ways that map directly to how a real mall operates and to concepts worth demonstrating academically:

1. Built for many shops, not one till

Every table — products, sales, stock — is shop-aware from day one, so the same system that runs one counter can aggregate revenue across the entire mall. Most comparable projects hardcode a single-shop assumption that has to be retrofitted later.

2. Data integrity as a design principle, not an afterthought

All billing operations are wrapped in JDBC transactions with explicit commit/rollback, so a crash mid-sale can never silently corrupt stock counts — a failure mode common in simpler POS clones.

3. Management-facing analytics, not just a receipt printer

A live dashboard (JFreeChart-driven) surfaces best-sellers, slow-moving stock, and per-counter performance — turning the system from a billing tool into a decision-support tool for the supervisor/owner.

4. Full audit trail

Every void, discount override, and stock adjustment is logged with who/when/why — a feature enterprise retail software charges a premium for, and a natural talking point for a viva/defense on accountability and security.

How It Compares

Capability	Typical Student POS Project	This Proposal (SMSMS)
Shop scope	Single shop assumed	Multi-shop / multi-counter from the schema up
Roles	Single login or none	Admin / Manager / Cashier with enforced permissions
Stock & sale consistency	Sequential, best-effort updates	Atomic DB transactions (commit/rollback)
Reporting	Basic tabular list	Charts + exportable PDF/Excel management reports
Accountability	Rarely logged	Full audit log of sensitive actions

06 Scope of the Project

In Scope

- Product, category, and multi-shop inventory management
- POS billing with discounts, tax, multi-mode payment
- Returns/refunds linked to original invoices
- Customer and supplier management
- Employee records and role-based login
- Sales, stock, and profit/loss reporting with export
- Local database backup/restore

Out of Scope (Future Work)

- Cloud/multi-branch synchronization across malls
- Mobile companion app
- Online/e-commerce storefront integration
- SMS/email invoice delivery (proposed as extension)
- Third-party payment gateway integration

07 Development Methodology

The project will follow an incremental/Agile approach, delivering the system in vertical slices (a working feature end-to-end) rather than horizontal layers, so a demoable build exists from an early milestone onward. Each increment includes design, implementation, and manual QA before moving to the next module.

08 Proposed System Architecture

The application follows a layered MVC-inspired architecture that keeps the Swing UI, business rules, and SQL access cleanly separated — improving testability and making the codebase defensible in a viva.

Layer	Responsibility
Presentation (Swing)	Login, POS, inventory, reports, and admin screens; input validation
Controller	Routes UI events to the correct service, manages navigation state
Service	Business rules — stock checks, discount/tax calculation, permission checks
DAO	All SQL via PreparedStatement; isolates persistence from business logic
MySQL Database	Normalized schema with transactional (InnoDB) tables

09 Proposed Timeline

Estimated over a 12-week development cycle, suitable for a single-semester PBL course:

Phase	Duration
Requirements & DB Design	Week 1 – 2
Core Setup & Authentication	Week 2 – 3
Inventory Module	Week 3 – 5
POS / Billing Module	Week 5 – 7.5
Returns, Customers, Suppliers	Week 7.5 – 9
Reports & Dashboard	Week 9 – 10.5
Testing & Polish	Week 10.5 – 12
Documentation & Defense Prep	Week 11 – 12

10 Tools & Technologies

Java 17+ Java Swing MySQL 8 JDBC HikariCP FlatLaf
JFreeChart iText / PDFBox Apache POI ZXing Maven jBCrypt

11 Feasibility Study

Aspect	Assessment
Technical	Java, Swing, and MySQL are mature, well-documented, and freely available; no specialized hardware required.
Economic	Entirely open-source toolchain — zero licensing cost, making it deployable on a mall's existing office PCs.
Operational	Runs locally without internet dependency, fitting malls with limited or unreliable connectivity.
Schedule	Modular scope fits comfortably within a 12-week PBL timeline.

12 Expected Outcomes

- ✓ A fully functional, installable desktop POS and inventory system tested against realistic mall sales scenarios.
- ✓ A normalized, transaction-safe MySQL schema documented with an ER diagram.
- ✓ A management dashboard demonstrating measurable insight (e.g., top 5 products, daily revenue trend) generated from live data.
- ✓ A written system report covering design decisions, testing, and evaluation — suitable for academic submission.

13 Future Scope

- Multi-branch synchronization for malls under the same ownership group
- Customer loyalty points and coupon redemption
- SMS/email e-receipts
- Cloud-based backup and remote reporting access

14 Conclusion

This project proposes more than a billing screen — it proposes a right-sized retail management platform built around the realities of a multi-shop mall: shared visibility, enforced accountability, and data that survives a crash mid-transaction. It is scoped to be fully achievable within a single PBL semester while still offering enough architectural depth — layered design, transactional integrity, role-based security, and analytics — to stand as a serious piece of engineering work worth defending in front of a supervisor or panel.

“The goal is not just a system that bills a customer — it's a system a mall manager could actually trust with their daily revenue.”

Signatures

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